

Nuoyan Wang

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EDUCATION

University of Illinois Urbana-Champaign

GPA: 4.0/4.0

Bachelor of Science in Computer Engineering

Aug 2021 – May 2024

Master of Science in Electrical and Computer Engineering

Aug 2024 – May 2026 (Expected)

Bronze Tablet Graduate (2024) – Top 3% of UIUC Grainger College of Engineering

Bradley A. Simmons Memorial Scholarship (2023) – Awarded to 5 students in ECE

EXPERIENCE

Graduate Teaching Assistant (ECE 385)

Aug 2024 – Present

Digital Systems Lab: University of Illinois Urbana Champaign

Champaign, IL

- Led 30 students by hosting quizzes and demo sessions weekly for RTL design on FPGA
- Graded demos/lab reports based on their adherence to standard industry documentation practices

Digital Signal Processing Engineering Intern + Co-op

May 2025 – Present

Shure Incorporated

Niles, IL

- Conducted 6 rounds of experiments to evaluate my DSP algorithms designs for multi-mic array
- Ensured my designs met considerations across various design aspects, collaborating with 15+ teammates from Acoustics, Embedded, Software, Test & Certification, FPGA Engineers, and Management
- Presented graphs/analysis in Python/MATLAB, with simulations in PyRoomAcoustics (python library)

Software Engineering Intern

June 2023 – August 2023

GoClouds

Beijing, China

Technology Research Intern

May 2023 – July 2023

Lenovo Group

Remote

PROJECTS

Master's Thesis: Chiplet Architecture-Aware Compiler for HLS | Vivado/Vitis HLS

Ongoing

- Professor Deming Chen, building on HIDA: A Hierarchical Dataflow Compiler for High-Level Synthesis
- Advanced compilers for HLS using Vivado/Vitis HLS, tested on multiple FPGAs that simulate chiplets
- Optimized workload partitioning across chiplets, targeting power/speed improvements in future multi-die SoCs

Out-of-Order CPU Competition | SystemVerilog, RISC-V, Verdi, Design Verification, Git

Apr 2024

- 1st place of 36 teams at UIUC, graded on 11 benchmarks like chip area, power usage, runtime, and IPC
- In a team of 3, designed and optimized a register-renaming out-of-order CPU in SystemVerilog
- Implemented early branch recovery, pipelined cache + daddda mul/div, BP, BTB, load-store queue

Convolutional Neural Net Accelerator on GPU | C, CUDA, Image Processing, AI/ML, Git

Nov 2023

- Utilized GPU benefits on CNN featuring variable strides, shared memory, streaming, and unrolling
- Placed Top 20 on competition leaderboard, maxing available optimization points

Unix-Based Operating System Design | C, Git

Mar 2023

- Created Linux-based OS, featuring interrupts, system calls, filesystems, scheduling, terminal, and signals

FPGA Music Synthesizer | SystemVerilog, Quartus, FPGA Design, Signal Processing

Dec 2022

- Created FPGA with RTL-based PWM audio, integrated analog DAC, 6-channel chord-enabled GPIO audio-out
- Achieved max 10/10 difficulty & A+, recognized at end-of-year showcase for outstanding difficulty and creativity

RELEVANT COURSEWORK

SoC	Computer Architecture	Parallel Programming	Digital Signal Processing	AI/ML
	Computer Vision	Operating Systems	Models + Algorithms	Data Structures

TECHNICAL SKILLS

Programming: Python, C, C++, CUDA, SystemVerilog, RISC-V, Intel x86, Java, JavaScript

Developer Tools: Git, AMD Vivado, Vitis, Vitis HLS, VSCode, VCS, Intel Quartus

Technologies: PyTorch, TensorFlow, Image Processing, Design Verification